

Decision and Social Choice

PE3101P — Lecture 8: Egalitarianism and prioritarianism

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Today

- 1 More on Harsanyi
- 2 Different Numbers of People
- 3 Egalitarianism
- 4 Prioritarianism
- 5 Ex Ante and Ex Post
- 6 Priority and Risk Aversion

Harsanyi's Aggregation Theorem

- ▶ Starts with some really plausible-looking principles.
- ▶ Derives a conclusion that is far from impossible to believe, but definitely controversial:

Utilitarianism

The best outcome is the one with the greatest sum of wellbeing*:

$$X \succsim Y \quad \text{iff} \quad w_1(X) + \cdots + w_n(X) \geq w_1(Y) + \cdots + w_n(Y),$$

where $w_i(X)$ is person i 's wellbeing in outcome X .

*Strictly, a sum of utilities linked to wellbeing: VNM utilitarianism.

Decision Theory and Social Choice

Decision theory

	Heads	Tails
Option A	5	5
Option B	11	0

Utilities, fair coin.

Social choice

	Alex	Darius
Option A	5	5
Option B	11	0

Wellbeing levels.

- ▶ Decision theory deals with risk: an option can be better in some states of nature and worse in others.
- ▶ Social choice deals with trade-offs between people: an option can be better for some people and worse for others.
- ▶ States of nature play the role that people play. Statewise Dominance $\leftrightarrow$ Pareto; risk $\leftrightarrow$ inequality.

Decision Theory and Social Choice



A visual representation of the differences between decision theory and social choice theory.

Two Analogous Principles

Anonymity

If two outcomes have the same wellbeing levels, just held by different people, they are equally good.

Stochastic Equivalence

If two gambles give the same outcomes with the same probabilities, just in different states of nature, they are equally good.

- ▶ Structurally, these are the same principle: one rearranges people, the other rearranges states.
- ▶ That doesn't make them equally plausible.

A Two-Way Connection (1)

McCarthy, Mikkola and Thomas (2020). Their premises, roughly:

- ▶ **Posterior Anonymity:** two gambles are equally good if they give the same chances of each pattern of wellbeing levels, ignoring who has which. This does the work of Anonymity and Statewise Dominance together: it gives $X \sim \text{LE}(X)$.
- ▶ **Anteriority:** if everyone faces the same chances of each wellbeing level under two gambles, they are equally good.
- ▶ **Reduction to Prospects:** among gambles that guarantee everyone the same wellbeing as each other, rank as an individual would rank her chances.
- ▶ The last two are parts of Ex Ante Pareto, about equal goodness only. No completeness, continuity or expected utility.

A Two-Way Connection (2)

- ▶ $X \sim \text{LE}(X)$, by Posterior Anonymity.
- ▶ In $\text{LE}(X)$, everyone faces an equal chance of each level in X . By Anteriority, it's as good as everyone getting the *same* level, each level equally likely.
- ▶ By Reduction to Prospects, rank that as an individual would rank an equal chance of each level: an equal chance of being each person.

Their theorem, roughly

Society ranks outcomes and gambles as an individual would rank an equal chance of being each person. E.g. $(10, 0) \sim (5, 5)$ iff a 50–50 chance of 10 or 0 is as good for a person as 5 for sure.

- ▶ The individual ranking fixes the social ranking, and vice versa. Add expected utility: Harsanyi's conclusion.
- ▶ The riskless ranking alone doesn't fix the rest; that needs more, e.g. expected utility.

A Simple Case (1): UNEQUAL and EQUAL CHANCES

EQUAL CHANCES

	Heads	Tails
Alex	10	0
Darius	0	10

Equally good in every state (Heads is unchanged).

By Statewise Dominance, both ways round:

UNEQUAL $\sim$ EQUAL CHANCES.

From Pairwise Anonymity to Anonymity (1)

Pairwise Anonymity

If outcome Y can be obtained from outcome X by swapping the wellbeing levels of two people, then $X \sim Y$.

(1, 2, 3, 4, 5, 6)

$\sim$ (**3**, 2, **1**, 4, 5, 6)

$\sim$ (3, **6**, 1, 4, 5, **2**)

$\sim$ (3, 6, 1, **5**, **4**, 2)

$\sim$ (3, 6, 1, 5, **2**, **4**)

(1, 2, 3, 4, 5, 6) $\sim$ (3, 6, 1, 5, 2, 4)

From Pairwise Anonymity to Anonymity (2)

- ▶ Any rearrangement of n people can be made by swapping two at a time:
 - ▶ swap the level that should end up first into first place;
 - ▶ then the level that should end up second into second place;
 - ▶ and so on. At most $n - 1$ swaps.
- ▶ Each swap gives an equally good outcome, by Pairwise Anonymity.
- ▶ So, by Transitivity, any rearrangement is equally good: Anonymity.
- ▶ This works for any finite number of people. With infinitely many, it doesn't.

Two Anonymity Principles

- ▶ Write πX for the outcome obtained from X by a rearrangement π of who gets which wellbeing level.

Anonymity

For any rearrangement π : $\pi X \sim X$.

Permutation Invariance

For any rearrangement π : $X \succsim Y$ iff $\pi X \succsim \pi Y$.

- ▶ Permutation Invariance rearranges both outcomes in the same way. Swapping Alex and Darius: $(10, 0) \succsim (5, 5)$ iff $(0, 10) \succsim (5, 5)$.

Permutation Invariance: No Rearrangement Is Better

Write $\pi^2 X$ for $\pi(\pi X)$: π applied twice. Likewise $\pi^3 X$, and so on.

Claim

Given Permutation Invariance and Transitivity, $\pi X \succ X$ is impossible.

- ▶ Suppose, for contradiction, that $\pi X \succ X$.
- ▶ Permutation Invariance (apply π to both): $\pi^2 X \succ \pi X$. Again: $\pi^3 X \succ \pi^2 X$. And so on.
- ▶ For some n , $\pi^n X = X$. (Every rearrangement of finitely many people, repeated enough times, gets back to the start: the cycle decomposition theorem.)
- ▶ Transitivity: $X = \pi^n X \succ \pi^{n-1} X \succ \dots \succ \pi X \succ X$. So $X \succ X$: contradiction.

With Completeness: Anonymity

- ▶ By the claim, $\pi X \not\sim X$.
- ▶ Also $X \not\sim \pi X$: apply the claim to πX and the rearrangement that undoes π .
- ▶ Completeness: either $X \succsim \pi X$ or $\pi X \succsim X$.
- ▶ Since neither is better than the other, both hold: $X \sim \pi X$. That's Anonymity.
- ▶ So the two principles differ only where there is incomparability.

Equally Good, or Incomparable?

- ▶ Permutation Invariance allows two verdicts about $(10, 0)$ and $(0, 10)$:
 - ▶ equally good: that's Anonymity;
 - ▶ *incomparable*: neither better, nor worse, nor equally good.
- ▶ If they're equally good, then $(11, 0) \succ (10, 0) \sim (0, 10)$: Alex getting 11 is better than Darius getting 10.
- ▶ That's not obvious. Incomparability can survive small improvements: $(11, 0)$ might still be neither better nor worse than $(0, 10)$.

Keeping Permutation Invariance, Dropping Anonymity

- ▶ In the simple case, Anonymity and Statewise Dominance gave $\text{UNEQUAL} \sim \text{EQUAL CHANCES}$.
- ▶ If $(10, 0)$ and $(0, 10)$ are incomparable, Statewise Dominance no longer gives that.
- ▶ So could EQUAL CHANCES be better than UNEQUAL , as the fairness intuition says?
- ▶ It would be better than $(10, 0)$ for sure, even though in each state its outcome is $(10, 0)$ or $(0, 10)$, neither of which is better than $(10, 0)$.

This Violates Negative Dominance

Negative Dominance

A gamble can't be better than another unless at least some outcome of the first is better than at least some outcome of the second.

- ▶ EQUAL CHANCES would be better than $(10, 0)$ for sure, though none of its outcomes is better than $(10, 0)$.
- ▶ Many people find Negative Dominance quite plausible.
- ▶ So perhaps one could avoid the theorem's conclusion by dropping Anonymity. But it's not so clear that that's a good way of doing it.

Ex Ante Pareto, One Person at a Time

EQUAL		
	Heads	Tails
Alex	5	5
Darius	5	5

Each step changed only one person's gamble. So, with Transitivity, we only need Ex Ante Pareto for changes to one person at a time.

Different Numbers of People

- ▶ What if the number of people can vary?

Neutral Addition

Adding people whose lives are neutral (wellbeing 0) leaves an outcome equally good.

- ▶ Add Neutral Addition to the theorem's premises, and we get total utilitarianism: $X \succsim Y$ iff X has at least as great total wellbeing as Y , whatever the numbers of people.
- ▶ Why: concentrate each outcome's wellbeing in one person, then remove the neutral lives. Next slide.
- ▶ Zero: 0 is the level where a life would be neither good nor bad. That fixes which lives are neutral.

Concentrate, Then Remove the Neutral Lives

A dash: that person doesn't exist.

	Alex	Darius	Philip	Total	Why
X	3	1	2	6	
$\sim$	6	0	0	6	The theorem
$\sim$	6	—	—	6	Neutral Addition
Y	2	5	—	7	
$\sim$	0	7	—	7	The theorem
$\sim$	—	7	—	7	Neutral Addition
$\sim$	7	—	—	7	Anonymity across different people

$(7, -, -) \succ (6, -, -)$: Alex at 7 is better than Alex at 6.

So, by Transitivity, $Y \succ X$: the greater total wins.

Egalitarianism

Egalitarianism (deontic)

Inequality should be avoided.

Egalitarianism (telic)

It is in itself bad if some people are worse off than others.

- ▶ In principle, our social ordering $\succsim$ could be about either: which outcome *ought to be chosen*, or which outcome is *better*.
- ▶ I'll mostly have the betterness interpretation in mind.

Egalitarianism at Work

	Alex	Darius	Total
UNEQUAL	10	0	10
EQUAL	5	5	10

- ▶ Utilitarianism: equally good. Same total.
- ▶ Telic egalitarianism: UNEQUAL is worse. Darius is worse off than Alex, and that is bad in itself.
- ▶ Deontic egalitarianism: UNEQUAL should not be chosen.

Equality for Its Own Sake

- ▶ If we claim that equality is good, we may mean only that it has good effects. Inequality can produce conflict, damage the self-respect of those who are worst off, or put some people in the power of others.
- ▶ Then equality has *instrumental* value: it is good as a means.
- ▶ For true egalitarians, equality has *intrinsic* (or *final*) value: it is in itself good.
- ▶ If inequality is in itself bad, it is bad even when it has no bad effects.

The Divided World

- ▶ The two halves of the world's population don't know about each other.

	One half	Other half	Average
(1)	100	200	150
(2)	145	145	145

- ▶ (1) has more wellbeing. Many people think (2) is better.
- ▶ The halves are unconnected, so the inequality in (1) has no effects. If (2) is better, inequality must be bad in itself: the telic view.

Inequality That Can't Be Prevented

- ▶ Some inequality can't in practice be prevented. Some people have lower expected lifespans than others; some are unlucky and catch diseases.
- ▶ Deontic egalitarianism: if the inequality can't be prevented, there is no injustice in it. Except for its bad effects, there is nothing to regret.
- ▶ Telic egalitarianism (on many versions): it is still in itself bad.
- ▶ For Parfit, cases like these most clearly separate the two views.

Levelling Down

	Alex	Darius
Before	10	5
After	5	5

- ▶ Nobody gains. Alex loses 5.
- ▶ Telic egalitarianism: the change is in one way good, since it removes the inequality — even if it's worse all things considered.
- ▶ The Levelling Down Objection: there is nothing good about it at all. It's not just that levelling down would be wrong: it isn't good in any way.
- ▶ Deontic egalitarianism avoids the objection: it needn't say the change is good in any way.

The Priority View

The Priority View (Parfit 1997: 213)

Benefiting people matters more the worse off these people are.

- ▶ For a utilitarian, how much a benefit matters depends only on its size.
- ▶ For a prioritarian, it also depends on how well off the person who gets it is.
- ▶ The priority isn't absolute: a big enough benefit to someone better off can outweigh a small benefit to someone worse off.

Not a View About Equality

- ▶ Egalitarians care about *relations*: how each person's level compares with others'.
- ▶ Prioritarians care only about *absolute levels*: how well off each person is.
- ▶ Parfit's comparison: at high altitude it's harder to breathe. That's because of how high you are, not because others are lower down.
- ▶ One can be both. But a prioritarian as such doesn't think inequality is bad in itself.

Two Reasons to Prefer EQUAL

	Alex	Darius
UNEQUAL	10	0
EQUAL	5	5

- ▶ Egalitarian: in UNEQUAL, Darius is worse off *than Alex*. That is bad in itself.
- ▶ Prioritarian: in UNEQUAL, Darius is *badly off*, so benefits to him matter more, whatever Alex has.
- ▶ Prioritarians favour equality only as a by-product of giving priority to the worse off.
- ▶ A utilitarian can also prefer equal *resources*: extra resources do less for someone who already has a lot. But not equal wellbeing.

Priority and Levelling Down

- ▶ Levelling down: Alex goes from 10 to 5, Darius stays at 5.
- ▶ Prioritarianism: that's worse, and in no way better. Nobody's absolute level goes up.
- ▶ So prioritarianism avoids the Levelling Down Objection.
- ▶ And it can still say (2) is better in the Divided World: the better off lose more than the worse off gain, but the worse off's gains matter more.

Separability over Persons

Separability over Persons

How two outcomes compare doesn't depend on the wellbeing of people who are equally well off in both.

- ▶ Prioritarianism has it built in: how much a benefit to someone matters depends only on how well off they are, whatever anyone else has.
- ▶ Egalitarianism needn't satisfy it: whether a change reduces inequality can depend on where the unaffected people are. Some egalitarian views happen to satisfy it; others don't.

Separability over Persons: An Example

	Alex	Darius	Philip
X	4	36	49
Y	9	16	49
X'	4	36	1
Y'	9	16	1

- ▶ Both times, going from X to Y , Alex gains 5 and Darius loses 20. Only Philip, who is unaffected, differs.
- ▶ Prioritarianism with $f = \sqrt{\cdot}$: X beats Y , and X' beats Y' . Philip's term cancels: $2 + 6 = 8$ against $3 + 4 = 7$.
- ▶ An egalitarian view could rank Y above X (Alex is the worst off) but X' above Y' (now Philip is).

A Representation Theorem for Prioritarianism

Adler and Holtug (2019). Fixed population. Suppose the ranking of outcomes satisfies:

- ▶ Strong Pareto, Pareto Indifference and Anonymity;
- ▶ Pigou–Dalton: a transfer from someone better off to someone worse off, which doesn't swap their ranks, is an improvement;
- ▶ Separability over Persons;
- ▶ Continuity: small enough changes to an outcome don't reverse a strict ranking.

Then

It ranks outcomes by $g(w_1) + \dots + g(w_n)$, for some strictly increasing, strictly concave g . And every such ranking satisfies all of the above.

Writing Prioritarianism Down

Prioritarianism

$$X \succsim Y \quad \text{iff} \quad \sum_i f(w_i(X)) \geq \sum_i f(w_i(Y)),$$

where f is increasing and strictly concave.

- ▶ Strictly concave: each extra unit of wellbeing adds less to how good the outcome is than the one before. So a benefit counts for more, overall, when it goes to someone worse off.
- ▶ For example, $f(w) = \sqrt{w}$: not the true one, just a convenient example for non-negative wellbeing.
 - ▶ (25, 25): $\sqrt{25} + \sqrt{25} = 5 + 5 = 10$.
 - ▶ (49, 1): $\sqrt{49} + \sqrt{1} = 7 + 1 = 8$.
- ▶ So (25, 25) is better than (49, 1), though the totals are the same.

Prioritarianism and Utilitarianism Can Disagree

	Alex	Darius	Total	Priority-weighted total ($f(w) = \sqrt{w}$)
X	25	0	25	$5 + 0 = 5$
Y	9	9	18	$3 + 3 = 6$

- ▶ Utilitarianism: X is better. It has the greater total.
- ▶ Prioritarianism with $f(w) = \sqrt{w}$: Y is better.
- ▶ Going from Y to X, Alex gains 16 and Darius loses 9.
Darius's loss counts for more, because he ends up so badly off.

Ex Ante and Ex Post Prioritarianism

- ▶ For gambles, there are two natural ways to apply the priority (there are others). Here, each person's betterness is risk neutral about wellbeing: utility = wellbeing.

Ex Ante Prioritarianism

Work out each person's expected wellbeing. Then take the sum of f of those.

Ex Post Prioritarianism

Work out the sum of $f(\text{wellbeing})$ in each state. Then take the expectation.

- ▶ The more common one is ex post prioritarianism.
- ▶ Square numbers from here: (49, 1) for sure; a coin toss giving (49, 1) or (1, 49); and (25, 25) for sure.

Ex Ante Prioritarianism at Work

Coin toss		
	Heads	Tails
Alex	49	1
Darius	1	49

- ▶ Each person's expected wellbeing: $\frac{1}{2} \cdot 49 + \frac{1}{2} \cdot 1 = 25$.
- ▶ Sum of f of those: $\sqrt{25} + \sqrt{25} = 5 + 5 = 10$. (49,1) for sure scores $\sqrt{49} + \sqrt{1} = 7 + 1 = 8$.
- ▶ So the coin toss is better than (49,1) for sure, though it is no better in either state. Ex ante prioritarianism violates Statewise Dominance: Diamond's exit.

Ex Post Prioritarianism at Work

Coin toss

	Heads	Tails
Alex	49	1
Darius	1	49

- ▶ Sum of $f(\text{wellbeing})$ if Heads: $\sqrt{49} + \sqrt{1} = 7 + 1 = 8$. If Tails: $\sqrt{1} + \sqrt{49} = 8$. Expectation: 8.
- ▶ (25, 25) for sure scores $5 + 5 = 10$ in every state.
- ▶ So (25, 25) is better than the coin toss, though each person's expected wellbeing is 25 either way. Ex post prioritarianism violates Ex Ante Pareto: the Full Information exit.

The Two Versions at Work

	(49, 1) for sure	Coin toss	(25, 25) for sure
Ex ante	8	10	10
Ex post	8	8	10

With $f(w) = \sqrt{w}$.

- ▶ Ex ante: the coin toss is better than (49, 1) for sure, though it is no better in either state. That goes against Statewise Dominance.
- ▶ Ex post: (25, 25) is better than the coin toss, though both give each person expected wellbeing 25. That goes against Ex Ante Pareto.
- ▶ So each version blocks a different move in the proof.

Ex Post Prioritarianism with One Person

	Heads	Tails
Sure thing	25	25
Gamble	49	1

Only one person exists. Her betterness is risk neutral about wellbeing.

- ▶ Expected wellbeing 25 either way: equally good for her.
- ▶ Ex post prioritarianism: sure thing $\sqrt{25} = 5$; gamble $\frac{1}{2} \cdot 7 + \frac{1}{2} \cdot 1 = 4$. The sure thing is better.
- ▶ So it violates Ex Ante Pareto even with nobody else around: no inequality, no unfairness.

Ex Ante and Ex Post Inequality

- ▶ *Ex post* inequality: in some state, people end up with different wellbeing.
- ▶ *Ex ante* inequality: people face gambles with different expected wellbeing.

	Heads	Tails	Ex ante equal?	Ex post equal?
UNEQUAL	(10, 0)	(10, 0)	No	No
EQUAL CHANCES	(10, 0)	(0, 10)	Yes	No
EQUAL	(5, 5)	(5, 5)	Yes	Yes

- ▶ EQUAL CHANCES: each person's expected wellbeing is 5, but whichever way the coin lands, one of them gets 10 and the other 0.

Ex Ante Equality, Any Ex Post Inequality

- Take any outcome, say (1, 2, 3). Spread it over three equally likely states, rotating who gets what (last week's lottery-equalisation):

	State 1	State 2	State 3
Alex	2	3	1
Darius	3	1	2
Philip	1	2	3

Each state has probability $\frac{1}{3}$.

- Everyone's expected wellbeing is 2: ex ante equality.
- But every state is as unequal as (1, 2, 3). This works for any outcome: ex ante equality is compatible with any amount of ex post inequality.

Ex Post Equality Guarantees Ex Ante Equality

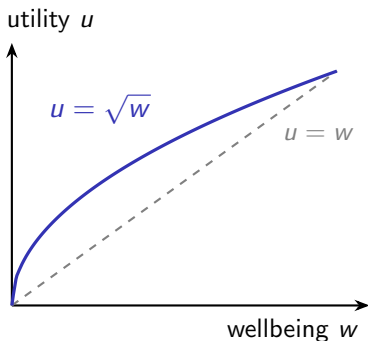
- ▶ Suppose every state is perfectly equal: in each state, everyone has the same wellbeing as everyone else.
- ▶ Then everyone faces the very same gamble. So everyone has the same expected wellbeing: ex ante equality.
- ▶ Not the other way round: EQUAL CHANCES is ex ante equal but unequal in every state.
- ▶ So guaranteed ex post equality is the more demanding kind (with a fixed population).

Ex Ante and Ex Post Egalitarianism

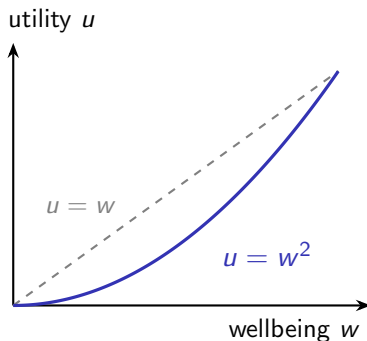
- ▶ Egalitarians face the same choice: inequality in people's chances, or inequality in outcomes?
- ▶ Ex ante egalitarianism: EQUAL CHANCES gives Alex and Darius the same chances (each has a 50–50 chance of 10 or 0), so it's better than UNEQUAL. That's Diamond's judgement, against Statewise Dominance.
- ▶ Ex post egalitarianism: EQUAL CHANCES guarantees an unequal outcome, so it's worse than EQUAL. That's the objection to Ex Ante Pareto.
- ▶ These aren't exclusive. An egalitarian can care about both ex ante and ex post inequality; so can a prioritarian.

Utility as a Function of Wellbeing

Two ways utility could curve against wellbeing.
The dashed line, $u = w$, is for comparison.



Concave:
risk averse about wellbeing



Convex:
risk seeking about wellbeing

Risk-Averse Betterness

- ▶ Now suppose instead that everyone's utility is $\sqrt{\text{wellbeing}}$.

Wellbeing	0	4	16
Utility	0	2	4

- ▶ A 50–50 chance of 16 or 0: expected utility $\frac{1}{2} \cdot 4 + \frac{1}{2} \cdot 0 = 2$.
- ▶ That's the utility of 4 for sure. So for each person, the gamble is exactly as good as 4 for sure.

The Proof Again, with Risk-Averse Betterness

Everyone's utility is $\sqrt{\text{wellbeing}}$. Fair coin.

	Heads	Tails	Why
(16, 0) for sure	(16, 0)	(16, 0)	Anonymity, Statewise Dominance
~ coin toss	(16, 0)	(0, 16)	
~ (4, 4) for sure	(4, 4)	(4, 4)	Expected Utility, Ex Ante Pareto

By Transitivity: $(16, 0) \sim (4, 4)$.

The Same Steps, with Prioritarianism

Now everyone's utility is wellbeing, and we use ex post prioritarianism with $f(w) = \sqrt{w}$.

Priority-weighted total in each state:

	Heads	Tails	Expectation
(16, 0) for sure	$4 + 0 = 4$	$4 + 0 = 4$	4
coin toss	$4 + 0 = 4$	$0 + 4 = 4$	4
(4, 4) for sure	$2 + 2 = 4$	$2 + 2 = 4$	4

All equally good: exactly the theorem's verdicts. Risk costs a person exactly what inequality costs overall.

Priority Equals Risk Aversion

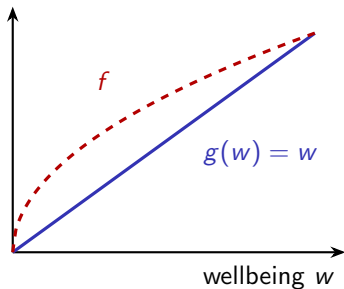
- ▶ In the theorem, the individual utilities do two jobs:
 - ▶ their *expectation* says what is better for each person;
 - ▶ their *sum* says what is better overall.
- ▶ Suppose everyone's utility is the same strictly concave function of wellbeing: utility = $f(\text{wellbeing})$.
- ▶ Then the theorem's conclusion is ex post prioritarianism with that f : gambles are ranked by the expectation of $f(w_1) + \dots + f(w_n)$.
- ▶ So, given the premises, the priority given to the worse off must be exactly the degree of risk aversion about wellbeing.

When Prioritarianism Disagrees with the Theorem

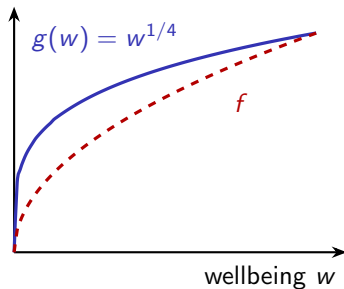
- ▶ Say everyone's utility is g (wellbeing). The theorem then gives ex post prioritarianism with $f = g$.
- ▶ A prioritarian whose f isn't just a stretched or shifted g disagrees, and has to reject a premise.
- ▶ Her f could be more concave than g : priority beyond risk aversion.
- ▶ Or less concave: less priority than risk aversion, though still priority to the worse off in wellbeing.

Same Priority Weighting, Different Utilities

Solid: utility g . Dashed: priority weighting $f(w) = \sqrt{w}$.



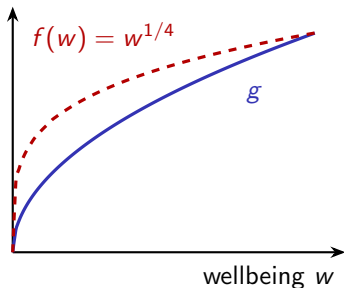
f more concave than g :
more priority than risk aversion



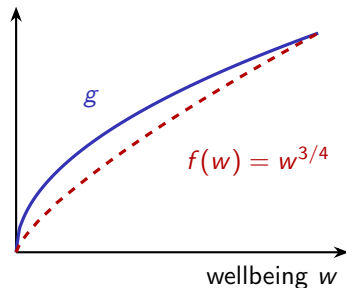
f less concave than g :
less priority than risk aversion

Same Utility, Different Priority Weightings

Solid: utility $g(w) = \sqrt{w}$. Dashed: priority weighting f .



f more concave than g :
more priority than risk aversion



f less concave than g :
less priority than risk aversion

Prioritarianism in Terms of Utilities

- ▶ Utility is $u = g(w)$, so wellbeing is $w = g^{-1}(u)$. So $f(w) = f(g^{-1}(u)) = (f \circ g^{-1})(u)$.
- ▶ The prioritarian's total, in utilities:
 $(f \circ g^{-1})(u_1) + \cdots + (f \circ g^{-1})(u_n)$.
- ▶ f is more concave than g exactly when $f \circ g^{-1}$ is concave.
- ▶ Example: $g(w) = \sqrt{w}$, so $w = u^2$.
 - ▶ $f(w) = w^{1/4}$: $(f \circ g^{-1})(u) = (u^2)^{1/4} = \sqrt{u}$: concave.
 - ▶ $f(w) = w^{3/4}$: $(f \circ g^{-1})(u) = (u^2)^{3/4} = u^{3/2}$: convex.
- ▶ So “priority to the worse off” means different things applied to wellbeing and applied to utilities.

Egalitarianism and the Theorem

- ▶ Whatever the risk attitude, the theorem's conclusion is a sum: one term for each person. So it is separable over persons.
- ▶ So an egalitarian view that isn't separable over persons has to reject a premise.
- ▶ Last week's two objections were about how people fare relative to each other: Darius's chance compared with Alex's; one person sure to end up worse off than the other.

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